



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST

AIR FLOW TESTER

MODEL AF-2400

ERROR AND TROUBLESHOOTING MANUAL

1 Error Indications

The tester indicates errors by showing error messages on the LCD display or blinking LEDs if any failures occur during power-on sequence, testing, calibration or compensation sampling. Concurrently, the tester outputs error signals.

1.1 Error Messages Displayed at Power-on and Remedial Actions

1) ROM and RAM checks

When the ROM and/or RAM is found defective, the HI NO GO and LO NO GO LEDs come on and further operations are infeasible. Ask the manufacturer for repair.

2) Error #5 "FS Offset Exceeding Limit"

Out of flow sensor offset tolerance limits (beyond $\pm 10\%$ F.S.). The flow sensor offset is abnormal. Ask the manufacturer for repair. Note that entering the FLOW SENSOR OFFSET screen in the CALIB mode temporarily cancels this error.

3) Error #6 "PS Offset Exceeding Limit"

Out of P sensor offset tolerance limits (beyond $\pm 2\%$ F.S.). The P sensor offset is abnormal. Ask the manufacturer for repair. Note that entering the P SENSOR OFFSET screen in the CALIB mode temporarily cancels this error.

The tester checks these items at the start of every test.

1.2 Error Messages and Remedial Actions

ERROR #	Message	Description	Probable Cause and Suggested Remedial Actions
1	Exceeding M.COMP Limit	Compensation limit overrange error	The compensation value has exceeded M.COMP LIMIT during compensation sampling. To decrease the compensation value, extend the CHG timer setting to reduce values measured in the DET2 stage.
2	A/D Converter Malfunction!	A/D converter malfunction	Replace the A/D converter. Ask the manufacturer for repair.
5	FS Offset Exceeding Limit	Flow sensor offset tolerance limit overrange (beyond $\pm 10\%$ FS)	The flow sensor offset has been found to be out of the tolerance limits at power-on. Ask the manufacturer for repair. Note that entering the FLOW SENSOR OFFSET screen in the CALIB mode temporarily cancels this error.
6	PS Offset Exceeding Limit	P sensor offset tolerance limit overrange (beyond $\pm 2\%$ FS)	The P sensor offset has been found to be out of the tolerance limits at power-on. Ask the manufacturer for repair. Note that entering the P SENSOR OFFSET screen in the CALIB mode temporarily cancels this error.

8	Improper Test Pressure	Test pressure drop	The test pressure has been found to be out of the preset test pressure limits when the CHG timer has expired. The probable cause is an improper P.Lo (P.Hi for vacuum models) test pressure limit setting or P sensor failure.
10	PS output saturated	Saturated test pressure sensor output	The P sensor output during testing has exceeded the tolerance limits. Check the P sensor offset and test pressure setting.
15	Air Valve Inactive.C2	No flows in BLW stage	The probable cause is degraded flow sensor sensitivity, pilot pressure drop, test pressure drop and/or inadequate BLW time setting. Recheck the BLOW CHECK LIMIT setting.
17	P.Lo(Hi) value is 0(Zero)	Test pressure limit set to 0	The P.Lo (P.Hi for vacuum models) test pressure limit has been set to 000.000.
18	RTC Battery Discharge	Low battery voltage	The built-in battery is weak. Check the battery voltage and, if necessary, ask the manufacturer for repair.

1.3 Errors Displayed when Tester is Active

Stage Symbol	Stage Name	Message	Cause and Remedial Actions
WAIT	Wait		
DL1	Delay		
(FCHK)	F check		
(PCHG)	Preliminary		
CHG	Charge		
(LDET)	Large flow detection		
SDET	Detection	The -000 reading blinks and Lo No Go LED comes on.	The tester has detected a reverse flow in the flow sensor when the stage is timed out. Disconnect the air source and adjust the sensor offsets. If the pre-charge pressure setting is too high, decrease it.
(NR)	Noise reduction		
BLW	Air Blow		
(EXH)	Exhaust		
END	End		

2 Points to Check when NO-GO Failures Occur in Succession

Error/NG Evaluation	Probable Cause		Suggested Remedial Actions
The HI NO GO workpiece failure occurs frequently.	The cause lies in the workpiece.	Continuous failures due to blow-hole leaks, etc. Leaks from sealed locations due to defective machining.	Check for leaks by visual air bubble tests using soap water or visual submersion air bubble tests.
		Pressure fluctuations due to workpiece deformation or volumetric changes in internal sealings.	Apply the preliminary charge and preliminary exhaust stages.
		Pressure fluctuations due to temperature changes.	Maintain workpieces at room temperature.
	The cause lies in the sealing fixture or piping.	Leaks due to deteriorated sealing rubber.	After checking the sealing rubber for wear or stuck foreign matter, perform tests using a conforming workpiece. Clean or replace the sealing rubber as necessary.
		Leaks from the piping, couplers or valves.	Check for leaks by visual air bubble tests using soap water.
		Effects of volumetric changes in the sealed locations, such as changes in clamping pressure.	Implement exhaust interference countermeasures to guard against effects from other testers, if any. Review the sealing fixture and rubber material design.
	The cause lies in the tester itself.	Leaks inside the tester. Sensor failure due to the ingress of water, oil, dust or any other foreign matter into the tester.	Perform a leaktight test with the WORK port valve of the tester closed.
		Improper timer settings, test pressure settings or coefficient settings.	Reset the tester settings to the factory-set defaults and perform a leaktight test.
	The cause lies in the compensation process.	The compensation value is improper because the workpiece used is leaking, deformed or affected by temperature.	Refer to the compensation graph and change the timer settings.
		The workpiece used for compensation sampling is of a different type.	Use a correct type of workpiece.
As soon as the CHG stage finishes, the HI NO GO LED blinks and the test is forced to end.	The P.Hi/LO test pressure monitoring setting is improper.		Check the P.Hi/Lo settings and change them, if necessary.
	Large flow (Constantly monitored by the P sensor)		After checking the sealing rubber for wear or stuck foreign matter, check the <i>flow rate</i> using a conforming workpiece.
	Test pressure failure		Check the pressure source and regulator settings.